was then measured at 19 foot-pounds. For the remaining measurements, the torque was only measured in the loosening direction. On bolts RA-1, RA-2, and RA-3, the position of the nuts were marked, the torque required to return the nut to its original position was also recorded. For bolts RA-1, RA-2, and RA-3, the movement torque in the loosening direction was 11.9 pound-force feet, 17.0 pound-force feet, and 22.4 pound-force feet, respectively. The torque to return to the original position measured 15.3 pound-force feet, 16.0 pound-force feet, and 21.8 pound-force feet, respectively. First movement torque measurements in the loosening direction for each of the attachment bolts is listed in table 1. According to the Piper Aircraft Maintenance Manual, the installation torque to be applied to the upper spar cap bolt heads and the lower spar cap nuts is 30 pound-force feet to 32.5 pound-force feet (360 pound-force inches to 390 pound-force inches).

After torque measurements were completed, the right main spar piece was disassembled from the center wing box assembly, and cuts were made to facilitate separating the mating fracture surfaces. Views of the inboard side of the fracture after completing the break with a lab fracture is shown in figure 15. The fracture surface had mostly rough matte-gray features consistent with ductile overstress fracture. River patterns associated with the fracture emanated from hole RD-1 located aft of the spar web consistent with fracture initiation near hole RD-1. The overstress fracture in the right main spar lower spar cap emanated forward, intersecting hole RC-1. As shown in figure 15, a reflective area was observed on the fracture surface at the lower aft side of hole RC-1.

The right wing main spar lower spar cap piece shown in figure 15 was further examined using an SEM, and resulting images of the reflective area at the forward side of hole RC-1 are shown in figure 16. At higher magnification, striations were observed consistent with fatigue. Fatigue features emanated from an origin area located at the hole surface as indicated with a bracket in the lower image in figure 16. A dashed line in the upper image in figure 16 indicates the fatigue boundary, extending up to 0.047 inch aft of the hole and 0.038 inch upward from the lower surface. Where the fatigue region intersected the lower surface, the fatigue crack length was 0.026 inch. No evidence of a preexisting crack was observed at hole RD-1 (where the overstress fracture of the spar cap initiated).

The overstress fracture at the forward side of the hole was located outboard of the transverse plane intersecting the hole center. A tilted view of the forward edge of the hole inboard of the overstress fracture is shown in the upper image in figure 17. As indicated, a crack was observed at the forward edge of the hole where it intersected the lower surface. A closer view of the crack fracture surface is shown in the lower image in figure 17. Smooth fracture features with curving crack arrest lines were observed consistent with fatigue fracture. Ratchet marks[2] were also observed, consistent with multiple fatigue origins. A bracket in the lower image in figure 17 indicates the fatigue origin area located in the chamfer between the hole bore and the lower surface of the main spar.

[2] A ratchet mark is a small step in the fracture surface formed when two adjacent fatigue cracks originate on slightly offset planes.